

EDITOR'S COMMENTS

Engaging with Practice for Impactful Research

By: **Susan Brown, Editor-in-Chief**
Stefan Seidel, Senior Editor
John D'Arcy, Senior Editor
Jungpil Hahn, Senior Editor
Carol Hsu, Senior Editor
Sabine Matook, Associate Editor
Maytal Saar-Tsechansky, Senior Editor
Maha Shaikh, Associate Editor
Jason Bennett Thatcher, Senior Editor

In this third editorial of *MIS Quarterly*'s 50th volume, we focus on the journal's long-standing relationship with practice. In the journal's early decades, engagement with practice was core, beginning with a separate applications section and interviews with CEOs to engage both research and practitioner audiences (Brown, 2026; Dickson, 1977, 1982). This tradition led to the IT issues studies (Brancheau & Wetherbe, 1987; Brancheau et al., 1996; Dickson et al., 1984; Niederman et al., 1991). This editorial is a nod to those studies and a reminder of the importance of this organizational connection to our research.

Information systems (IS) is a discipline that emerged from the intersection of computing and business. *MIS Quarterly* has reflected that intersection from its inception. We believe that research can and should leverage practitioner concerns as signals of phenomena that deserve deeper explanation. Such signals are most useful early, while organizational arrangements are still forming. With artificial intelligence (AI) above all, organizational arrangements are being set faster than we can observe them, from decision-making to operations to governance. Given this backdrop, we hope this editorial inspires renewed engagement with practice so that the problems we study and the recommendations we make can be better aligned with practitioner needs and interests.

Our Approach

We draw on two key sources of evidence for this editorial. Central to our approach is a multi-region, modified two-round Delphi study with senior IS executives, designed to identify the most critical IS-related issues organizations currently face as well as the developments they expect over the next 5 years.¹ We compare these results with the 2025 SIM IT Issues and Trends study (Johnson et al., 2026). Our editorial returns to the historical tradition of SIM key issues studies published first in *MIS Quarterly* and more recently in *MIS Quarterly Executive* (Brancheau & Wetherbe, 1987; Brancheau et al., 1996; Dickson et al., 1984; Johnson et al., 2026; Niederman et al., 1991). The early *MIS Quarterly* studies in this tradition also used multi-round Delphi designs.

In Round 1, 25 executives from five regions (North America, South America, Europe, Africa, and East Asia) openly nominated issues organizations currently face and developments they expect. We de-identified the responses before analysis. One researcher coded the responses manually. A second researcher coded them with the support of a large language model (ChatGPT 5.6 Sol, July 2026) using AI in a research-assistant role (Gopal et al., 2025) to surface candidates the first coder might have missed. We then reviewed, reformulated, and merged the candidates in a consensus meeting without model support, producing 15 issues and 18 developments.

Twelve executives completed Round 2. All 12 saw the consolidated Round 1 results, ranked both lists, and rated the importance of each item on a scale from "not important" to "among the most critical issues (developments)." Given the small panel, we report tiers rather than exact ranks. The first two tiers of current issues carry the strongest emphasis in our data, as does Tier 1 of the expected developments.

¹ The study received ethics approval from the University of Queensland ethics committee (approval code: 2026/HE011082).

Table 1. Current Issues by Tier

Tier 1	Cybersecurity capability in AI-amplified threat environment
	Data quality, access, governance, analytics foundations
	AI adoption and value realization
Tier 2	Technology leadership and change management
	Agentic AI integration and process redesign
	AI governance and responsible use
	Workforce skills and AI literacy
Tier 3	Service delivery
	Digital customer experience
	Digital resilience, continuity, ecosystem dependence
Tier 4	Legacy modernization, interoperability, technical debt
	Frontier technology risk
	Regulation and strategic autonomy
	Technology economics, TCO, funding, portfolio trade-offs
	Ecological and societal risks

Note: Tiers reflect visible breaks in average rankings within each list rather than estimates of population order. The tiers should not be interpreted as evidence of consensus.

Table 2. Expected Developments by Tier

Tier 1	Data, knowledge, and real-time intelligence foundations
	Talent and organizational AI capability
	Agentic and autonomous AI
	Cybersecurity threats and AI-enabled attack or defense
	Work and process redesign
Tier 2	Adaptive resilience
	Cloud, edge, composable architecture, and infrastructure
	AI-enabled process automation and software/IT delivery
	AI governance, auditability, and regulation
Tier 3	Digital identity
	Conversational, personalized, and omnichannel experiences
	Pervasive enterprise AI
	AI-enabled science
	Technological sovereignty, geopolitics, and strategic dependence
	Physical AI and robotics
	Sustainability and circular digital systems
	Societal and macroeconomic consequences
	Quantum and post-quantum technologies

Note: Tiers reflect visible breaks in average rankings within each list rather than estimates of population order. The tiers should not be interpreted as evidence of consensus.

The overlaps with the SIM study are substantial: AI, cybersecurity, and data-related issues rank near the top of both (Johnson et al., 2026). The difference is granularity. Where Johnson et al. (2026) treated AI as one category, our respondents separated it into distinct concerns, from value realization and governance to agentic integration and work redesign, and raised several longer-range developments that a single category does not capture. Johnson et al. noted the issue themselves: Digital transformation fell out of their top 10 for the first time since its introduction in 2017, which they link to the rise of the specific technologies that the category had long contained. This distinction matters because executives did not describe AI as one adoption problem. They described connected questions about data foundations, value, work redesign, governance, security, skills, and resilience. Compressing AI into a single category obscures how these problems interact and what future IS research needs to explain.

One other pattern is worth noting. Ecological and societal risks appeared as issues in Round 1 but fell into the last tier when executives ranked immediate concerns. We do not read that as evidence that these risks are unimportant. It may instead signal a gap between what organizations prioritize now and the broader consequences researchers need to anticipate. We return to that issue when considering how the benefits and costs of AI are distributed.

Drawing on these results, we focus on four problems that our team felt were consequential for IS research: cybersecurity in an AI-amplified threat environment, realizing value from AI, AI governance and responsible use, and digital resilience and business continuity. The first two were Tier 1 current issues. The other two fell outside the top tier of current issues and into the second tier of expected developments, linking present concerns to what respondents anticipate. We selected these four not because they are exhaustive of the results but because, together, they show how AI changes value creation, the allocation of authority, exposure to risk, and organizational dependence. We then considered two developments whose implications remain less settled to illustrate why engagement with practice matters at the early stages of a phenomenon.

Cybersecurity Capability in an AI-Amplified Threat Environment

Cybersecurity and AI have not always been at the forefront of the organizational IT agenda. Early SIM studies published in *MIS Quarterly* ranked both issues well below concerns such as strategic planning, systems development, and IT infrastructure, and by the mid-1990s both had been dropped from the list altogether (Brancheau & Wetherbe, 1987; Brancheau et al., 1996; Dickson et al., 1984; Niederman et al., 1991). Four decades later, the priorities have shifted dramatically. In the 2025 SIM IT Issues and Trends Study, AI ranked first among the issues most important to their organizations, followed by cybersecurity (Johnson et al., 2026).

Our Delphi study bears out this historical shift and suggests that AI and cybersecurity are increasingly converging. Cybersecurity capability in an AI-amplified threat environment was a Tier 1 current issue in our study. The qualitative responses help explain why. Executives described an environment in which AI is simultaneously changing attackers' capabilities, expanding and altering the organizational attack surface, and changing the capabilities organizations need to defend themselves.

On the offensive side, AI may lower barriers to sophisticated attacks while increasing the speed, scale, personalization, and adaptability of their development and execution. One respondent described "AI-driven attacks" occurring "at machine speed," highlighting how traditional defenses are increasingly inadequate. The threat is not solely external. Intelligent agents, automated workflows, and data-driven solutions expand the attack surface "not just in volume, but in complexity." AI compounds this problem because these systems require access to organizational data and connections to databases, APIs, cloud platforms, third parties, and other systems (Schneier & Vance, 2025).

Agentic AI is a case in point. AI differs from prior generations of organizational technologies partly through its increasing autonomy, learning capability, and ability to undertake tasks that previously required human decision-making (Berente et al., 2021). AI agents may receive identities, credentials, and access to sensitive data. They may also interact with other systems and agents, initiate transactions, generate or execute code, and take consequential actions with limited human intervention. Cybersecurity must therefore extend from securing human users and technological assets to securing nonhuman organizational actors that exercise delegated authority.

These developments raise questions about whether established conceptualizations of organizational cybersecurity capability translate to an AI-amplified threat environment. Identity and access management, endpoint protection, encryption, vulnerability management, employee training, governance, and incident response remain necessary. Our respondents suggest that these capabilities may increasingly represent a baseline rather than a sufficient response.

Across these accounts, AI appears as part of the threat and the defense. Attackers and defenders draw on similar technological capabilities, while organizational AI adoption creates additional assets and forms of agency that require protection. These insights point to three research opportunities.

From Practitioner Problem to Research Opportunity

The first opportunity concerns the coevolution of offensive and defensive cybersecurity capabilities (Ghasemaghaei, 2026). Much cybersecurity research asks whether particular technologies, controls, governance mechanisms, or behaviors reduce risk (Angst et al., 2017; Benaroch & Chernobai, 2017; Cram et al., 2019). An AI-amplified environment makes effectiveness more relational and dynamic because advances in defense invite corresponding adversarial adaptation. What enables organizations to learn faster than the attackers confronting them? When do AI-enabled defenses remain effective, and when do they trigger another cycle of adaptation? These questions make the relative rates of learning by attackers and defenders central to cybersecurity effectiveness.

The second opportunity concerns AI agents as organizational actors with cybersecurity consequences. IS cybersecurity research has focused largely on people who comply with policies, misuse systems, fall victim to social engineering, or identify threats (Chen et al., 2021; Cram et al., 2019; Jensen et al., 2017; Straub, 1990). AI agents may now occupy that position and possess credentials, access information, interact with other systems, and act with limited human intervention. How should organizations define the scope of an agent's authority? How can they distinguish legitimate but unexpected behavior from compromised behavior? Who is accountable when an authorized agent creates an incident, and what oversight remains feasible at machine speed? These questions also raise a new question: Should an autonomous agent with organizational credentials and delegated authority be treated as an insider?

The third opportunity concerns the speed mismatch between technological change and organizational adaptation. Respondents described threats operating at machine speed alongside pressure to deploy AI rapidly, while security capabilities struggle to keep pace. Organizations need time to recognize risks and build the capacity to respond (Angst et al., 2017). When does AI adoption outpace the ability to secure what is being deployed? What governance structures allow security considerations to develop alongside innovation rather than follow it? Why do some organizations adapt before vulnerabilities become evident while others respond afterward? These questions highlight how the pace of adoption—not only the controls—is a security issue in itself.

Collectively, these opportunities suggest that cybersecurity capability may depend less on possessing a fixed set of resources at a point in time than on sensing changing threats, learning, and reconfiguring those resources (Pavlou & El Sawy, 2011; Teece, 2007). Methodologically, this evolution calls for designs capable of capturing adaptation. Longitudinal studies of capability development, field studies accompanying AI and agentic-system deployments, simulations of attacker-defender adaptation, and collaborations that provide access to real-time security and incident-response data could reveal how cybersecurity capabilities evolve in practice. We are not the first to call for greater attention to time in cybersecurity research or for access to “real” data (Cram et al., 2024). These approaches become more important as the convergence of AI and cybersecurity accelerates the evolution of threats and defenses.

Realizing Value from AI

Although AI now occupies a prominent place on the organizational technology agenda, our respondents suggested that access to models and adoption alone are no longer the central challenge. Value realization was the other Tier 1 current issue in our study, and executives are increasingly concerned with whether, where, and how organizations can translate AI investments into realized value. One respondent described a “gap between expected AI adoption impact and actual performance improvement.” Respondents noted that initiatives often stall at the minimum-viable-product stage, increasing technical debt without capturing value, or produce isolated task gains that do not scale into durable organizational improvement.

Respondents described value realization from AI as an organizational transformation problem involving leadership, incentives, decision rights, skills, cross-functional coordination, data, architecture, governance, and costs. As one executive observed: “AI adoption is not about technology; it's about the capacity for management, decision-making, and effective application.” They were not simply asking for more or better general-purpose models but, for instance, wanted “productized” solutions that could produce value reliably.

Their comments identified four interdependent challenges: organizational adaptation, the joint design of AI and work, complementary data and technical foundations, and economic and ecosystem viability. The last includes immature vendor solutions and the potentially unsustainable costs of implementation, integration, governance, training, and ongoing operation. Emerging research points in a similar direction, suggesting that a transition from general-purpose AI to AI that is tailored to complement its context offers new opportunities to realize value beyond those that can be achieved with general-purpose AI capabilities.

The respondents' comments point to a joint design problem. AI capabilities, workflows, roles, processes, and human-AI interactions have to be designed and adapted together so that technical improvements translate into organizational performance. Organizations also need AI-ready data, integrated platforms, interoperable architectures, governance mechanisms, and economic models that account for integration, training, supervision, and ongoing use.

From Practitioner Problem to Research Opportunity

The first opportunity concerns the measurement of organizational value. AI's value is easiest to assess when model performance and decision outcomes can be measured directly. In many organizational settings, however, outcomes are delayed, noisy, heterogeneous, or difficult to attribute to AI. The problem is not only retrospective measurement. Organizations need prospective measures that guide AI design and decisions about where AI should be used (Gao et al., 2021). The business value of IT literature offers a foundation for linking technology, complementary resources, and performance (Kohli & Grover, 2008; Melville et al., 2004). AI raises additional challenges: We need frameworks to assess prospective and realized value at multiple levels, from individual tasks to roles and organizational value, in order to inform the design of both AI and processes and the continuing adaptation of each to the other. Such measures need not be perfect but should provide sufficiently reliable guidance.

The second opportunity concerns distribution. Measures also need to capture how AI benefits and costs are distributed: AI may create value for the organization while shifting costs to employees, consumers, the environment, or other stakeholders. In some settings, organizational, consumer, and societal value may reinforce one another; in others, organizational value may be created at their expense (Pang et al., 2024). Research should examine when these forms of value complement each other and when they conflict. Such work can inform system design, business models, policy, and regulation aimed at producing more broadly shared value.

The third opportunity concerns the joint design of AI and work. Research on human-AI collaboration in business has largely focused on how to design work, processes, and human engagement with AI to improve outcomes. Yet some constraints cannot be designed away, including human expertise, collaborative behavior, and workflows beyond the organization's control. An important opportunity is the development of context-specific AI collaborators that learn and account for these realities to more reliably produce better outcomes for a given work design and set of collaborators (Bansal et al., 2021; Wolczynski et al., 2024; Gao et al., 2021). Such context-aware (grounded) AI could reduce reliance on individuals' ability to leverage AI advantageously and more reliably add value in a given environment. Simultaneously, although value realization depends on cost-effective human-AI collaboration, this objective has received limited business scholarly attention (Wolczynski et al., 2024).

From an organizational perspective, AI may enter work without an employee making a separate adoption decision. A vendor may add it to an existing system, or an organization may redesign a process around it. The work design question concerns how tasks, authority, and judgment are divided between people and AI, not simply whether employees accept the system (Baird & Maruping, 2021). Human-AI performance depends on the task, the expertise, and advice-taking behavior of the human collaborator and on how work is allocated (Dell'Acqua et al., 2026; Fügner et al., 2022; Jussupow et al., 2021). Having general-purpose AI with superhuman capabilities does not necessarily produce better outcomes, relative to what can be achieved without AI. Research needs to identify process designs that enable people with different expertise to use, challenge, correct, and override AI in ways that reliably improve performance and examine how to design channels for contesting AI decisions as authority continues to shift after the initial delegation (Ostern et al., 2026). What counts as effective use when AI initiates action and employees mainly review or repair its output? When does delegation improve the next decision but weaken expertise or professional judgment over time? And what are the costs of using AI, from computation to the time people spend prompting, checking, considering AI's suggestions, correcting, and coordinating its work?

The fourth opportunity concerns scaling AI-based innovation. Respondents questioned how organizations can move beyond isolated pilots and decentralize AI-enabled innovation without duplicating models, data pipelines, and governance. Research can compare deployment models that balance reuse and centralized control against local experimentation, duplication, coordination, and control costs. It should also examine how organizations sequence investments and decide whether to expand, redesign, or terminate initiatives as evidence of value and recurring costs accumulates. The next frontier of organizational AI

is no longer adoption or access to more capable general-purpose models but the reliable creation of value. Taken together, these questions ask how organizations measure value, allocate work and authority, design context-aware AI, scale AI's integration, and account for who receives the benefits and bears the costs. They also direct attention to whether AI, work, and organizational arrangements can continue to adapt together as people, contexts, and costs increasingly differ.

AI Governance and Responsible Use

Our Delphi study indicates a tension between the attention AI governance commands and the priority it receives. In Round 1, the majority of respondents raised AI governance, responsible AI, governing agentic AI, or shadow AI, often in technically detailed and urgent terms. One respondent warned that agentic systems are moving into production “faster than governance frameworks can adapt.” Yet AI governance fell into Tier 2 in Round 2, received no first-place votes, and was rated “critical” or “among the most critical” by only three of 12 respondents. The SIM study shows a related pattern: AI became the leading issue, while “constraints from existing governance” ranked behind more immediate implementation barriers (Johnson et al., 2026).

The first concern is integrating and governing agentic AI within existing systems and operations. One respondent responsible for AI architecture in financial services explained that existing model-risk guidance “was designed for static, single-model deployments” and does not readily address multi-agent behavior, dynamic tool use, or autonomous decision chains. The same respondent noted that the difficulty is jurisdictional as much as technical because expectations differ substantially across the United States, the European Union, the United Kingdom, and Asia-Pacific jurisdictions.

The second concern is shadow AI: adoption and development outside or ahead of formal governance. One chief technology officer described initiatives that “crop up” as executives encourage employees and units to experiment, often without established processes and controls, and noted a lack of tools to detect this use. A chief information officer warned that shadow AI may produce false outcomes and harm customer or employee trust, while scalable approaches for managing end-user developed AI remain rare. The problem is therefore not simply unauthorized use. Executive pressure to experiment can create a gap between the speed of adoption and the capacity of official channels.

The third concern is the gap between governance on paper and governance in practice. One chief technology officer described controls that function “like a stage gate” and initially treat every use as high risk, rather than as an operational filter matched to the risk of a particular application. Another respondent emphasized the need to translate high-level requirements into “concrete, verifiable” organizational behavior. In some organizations, restricting access to a single vetted product serves as a proxy for governance capability while policy is still being developed.

The fourth concern is the practical immaturity of explainability and auditability for the systems governance is meant to oversee. One respondent described explainability tooling for large language models as “still primitive,” relative to traditional statistical models, and noted that many validation teams lack experience assessing emergent behavior, prompt sensitivity, or agent decision chains. Techniques developed to explain a prediction do not readily transfer to systems that plan, delegate, use tools, and act.

From Practitioner Problem to Research Opportunity

The first opportunity concerns distinguishing declared from demonstrated governance. Shadow IT research shows that unauthorized use can signal that formal governance does not meet legitimate user or business needs (Haag et al., 2022; Haag & Eckhardt, 2024). AI adds regulatory ambiguity and much lower barriers to informal adoption (Lanamäki et al., 2025). Research could examine when governance decouples from agent and employee behavior, whether operational indicators such as audit trails, override rates, and escalation frequency capture governance capability better than policy adoption, and whether the gap between declared and demonstrated governance predicts incidents or failed deployments (Bromley & Powell, 2012; Meyer & Rowan, 1977).

The second opportunity concerns accountability and oversight for agents embedded in existing systems. As the cybersecurity section noted, agents may possess identities, credentials, and delegated authority. Governance therefore begins with traceability: An agent's identity, purpose, permissions, and accountable principal must remain visible as the agent is updated or delegates further. Oversight cannot be a generic safeguard. Continuous monitoring, bounded autonomy, and post hoc review fit different levels of speed, reversibility, and consequence. Research should ask how organizations choose among these arrangements, set decision rights and escalation thresholds, and preserve the ability to contest agent actions (Grisold et al., 2025; Ostern et al., 2026; Rai et al., 2026).

The third opportunity concerns governance across unstable regulatory regimes (Butler et al., 2023). Requirements differ across jurisdictions and continue to change. Organizations operating across jurisdictions must adhere to requirements that differ from one another and that continue to change. A governance architecture built for one regime may not survive the next. Research should examine how flexible governance architectures should be designed and where governance authority should be located—centrally, in business units, or in the system itself. It should also ask how responsibility should be apportioned when developers, platform providers, and deploying organizations contribute to the same incident.

The fourth opportunity concerns explanation. Auditability establishes a record of what occurred; explanation concerns whether people can understand the basis and sequence of an AI action. Most techniques explain a single prediction or decision (Fernández-Loría et al., 2022). An agent's behavior unfolds through linked choices about planning, tool use, delegation, and stopping. Research should examine how to explain such trajectories, to whom explanations are owed, and what technical and human capability organizations need to interpret them (Rivera et al., 2025).

Finally, broader practitioner conversations raise a question that our panel, drawn largely from well-resourced organizations, could not resolve: how governance works when organizations lack dedicated legal, compliance, and technical functions. This is an editorial extension rather than a Delphi finding, resulting from a team member's interaction with practice. Research should examine whether shared infrastructure, sector-level intermediaries, or graduated obligations enable smaller organizations to govern AI without assuming that governance capability is readily available.

Taken together, these opportunities require research that observes governance in operation, not only in policy documents. Comparative studies that connect policies to system logs, audit trails, or agent-action records can test where decoupling occurs. Longitudinal work can follow organizations as AI capabilities and regulatory expectations change, while design science can develop mechanisms for identity, permissioning, auditability, contestability, explanation, and reversibility. The problem respondents described was not a shortage of frameworks; it was a shortage of governance that works at the speed and scale of the systems being governed.

Digital Resilience and Business Continuity

The SIM studies reveal two patterns that help explain the current concern with resilience. First, attention to business continuity appears to be disruption driven. It ranked relatively low before COVID-19, rose sharply during the pandemic, and then began to decline. Second, concern and investment remain misaligned. In 2025, disaster recovery and IT continuity planning ranked 25th among organizations' largest IT investments, but sixth among areas IT leaders believed should receive more investment and third among those they found most personally worrisome (Johnson et al., 2026). These patterns suggest that attention rises when disruption is visible but is harder to sustain when the benefits of preparedness take the form of losses that did not occur.

Our respondents support the growing awareness of the need to prepare for future disruptions, while also suggesting a shift from business continuity and recovery toward adaptive, cyber, and operational resilience. As one explained, the goal is to move beyond "firefighting," so the organization can "thrive through disruption." Third-party dependencies, economic interdependence, and AI make this harder because disruption can propagate across systems and organizations. The shift is therefore from recovery after an event toward a more proactive capacity to prepare, adapt, and continue operating.

Digital resilience refers to the capacity organizations build through digital technologies to absorb major shocks, adapt in response to those shocks, and transform into a new stable state (Boh et al., 2023). Much prior work examines digital technologies as resources for responding to nondigital shocks, such as a pandemic or natural disaster (Abbasi et al., 2024; Rai, 2020). Our respondents also described digital technologies, especially AI and interconnected infrastructures, as sources of disruption. This changes the problem from using digital technologies to recover from an external shock to building resilience into the systems on which organizations increasingly depend.

From Practitioner Problem to Research Opportunity

These problems are sociotechnical. Resilience depends on interactions among technologies, organizational routines, decision makers, partners, and institutions rather than on additional technology alone (Sarker et al., 2019). Studying one element in isolation risks an impoverished understanding and inadequate solutions. Four research opportunities follow.

The first opportunity concerns the resilience concern-investment gap. The sociotechnical ensemble includes IT leaders' risk assessments, executive attention, prior disruption experience, and the technologies that make dependencies visible. The tensions involve continuous preparedness versus short-term efficiency, visible returns versus avoided losses, and centralized accountability versus distributed dependence. Research should ask how IT leaders make resilience risks visible and economically justifiable, how executive framing shifts organizations toward reactive or proactive resilience, and how these decisions condition the capacity to absorb, adapt, and transform over time (Boh et al., 2023).

The second opportunity concerns collaborative and distributed AI architectures. Resilience increasingly depends on data, models, and infrastructure distributed across organizational boundaries. Federated learning and open models may allow organizations to collaborate without pooling all data, but they also create questions of control, visibility, and appropriation (Han et al., 2026; Nagle & Yue, 2025). Not all open models are equally open: access to weights, training data, source code, and the ability to modify or leave an arrangement vary. The sociotechnical ensemble includes models, data, infrastructures, contracts, licenses, community norms, orchestrators, and trust relationships. The tensions involve openness versus control, collaboration versus competition, privacy versus auditability, and decentralization versus orchestrator power. Research should examine how disparities in local data shape shared models, when orchestrators enable or constrain collective resilience, and how visibility across model ecosystems affects organizations' capacity to adapt.

The third opportunity concerns AI and autonomous agents. Agents may support resilience, but they may also act across system and organizational boundaries at speeds that limit human intervention (Grisold et al., 2025; Zhang et al., 2021). Here, the sociotechnical ensemble includes agent capabilities, human judgment, organizational routines, technical infrastructure, legacy systems, and professional norms. This creates tensions between autonomy and control, speed and oversight, automation and professional learning, and local agent action and system-wide consequences. Research should ask how control shifts during normal operations and disruptions, which forms of oversight remain effective at scale, how agent identities and permissions should be configured, and how adaptive agents can be integrated with legacy systems and organizational data without weakening continuity, learning, or professional judgment.

The fourth opportunity concerns digital sovereignty as a component of resilience (Anand et al., 2026; Floridi, 2020; Lehuédé, 2024). Organizations and states seek greater control over critical technologies while remaining dependent on global cloud, AI, semiconductor, identity, and software ecosystems. The relevant sociotechnical ensemble includes enterprise architecture, infrastructure, organizational capabilities, vendor relationships, and institutional conditions. Its central tensions involve sovereignty versus interdependence, resilience versus efficiency, portability versus lock-in, present value versus future optionality, and the risk of replacing one dependency with another. Research should examine how organizations identify strategic dependencies; when multi-cloud, hybrid, open-source, edge, federated, or redundant arrangements strengthen rather than complicate resilience; and how technological reversibility, credible exit options, and substitution paths can be preserved as technologies and geopolitical conditions change.²

These questions call for longitudinal and comparative research. Resilience develops through decisions made before, during, and after disruptions. Research should therefore trace how resilience priorities and investments change over time and avoid treating any one organizational or geographic model as universal (Schlackl et al., 2026; Venkatesh et al., 2016). More broadly, these questions are well suited to phenomenon-focused research (Monteiro et al., 2022; Ter Wal et al., 2026). Phenomenon-based theorizing neither privileges a method nor ignores current knowledge. It begins with puzzles that existing theory explains poorly and asks what becomes visible when researchers remain close to the particular features of a setting. That principle applies across all four topics.

Looking Ahead

The four preceding sections address problems that practitioners already clearly recognize. The responses also raised less-settled developments that could become important in the future. We suspect that their lower salience may reflect uncertainty about what they will mean for organizations rather than a lack of importance. We discuss two because they show why IS researchers need to engage with practitioners before technologies and sociotechnical systems become established.

² Europe's Tech Sovereignty package aims to reduce dependence on non-EU technology by strengthening European control of critical chips, cloud, AI, data and digital infrastructure—boosting security, resilience and competitiveness. See <https://digital-strategy.ec.europa.eu/en/policies/eu-tech-sovereignty>

Physical AI and robotics: Respondents described physical AI as a development worth watching. Existing IS research has examined robots in hospitals, public services, and other service or manufacturing settings (Mettler et al., 2017; Robert et al., 2024; Wang et al., 2024). One respondent pointed to humanoid robots used for safety work, where physical systems may protect equipment or human lives. Another expected physical AI to improve performance in industries that depend on physical labor.

A less settled question is whether increasingly capable embodied systems will move into knowledge-intensive and office work. Access to language and reasoning models may allow physical AI to perform a broader range of tasks, but organizational value will depend on more than technical capability. Readiness may include the design of physical environments, access to organizational data and knowledge, worker skills, procedures for interruption and override, and agreement about the roles such systems should occupy. IS research can examine where physical embodiment adds value, how work processes and physical environments must be redesigned, and how organizations determine whether a use case merits the additional security, governance, coordination, and maintenance it creates.

Several questions follow. How should organizations protect data and organizational knowledge when embodied agents can perceive, move, and act? How should authority, oversight, and accountability be allocated when AI combines digital agency with physical action? When does physical AI complement human judgment, and when does it narrow or displace it? Design science and action research can engage organizations while use cases are still being formed, and field studies can examine value in use rather than capability in the abstract. Trace data from system actions, environmental interactions, energy use, failures, and human intervention may also make it possible to study how these systems change through use and how organizations learn to work with them.

Future uncertainty: Respondents differed in how they anticipated advanced technologies, and their views appeared to vary with industry, location, and governance maturity. Given the small panel, these differences should be treated as observations rather than regional comparisons. Even so, several responses shared a concern about losing control as technologies become more autonomous and interconnected. Respondents referred to “systemic failure,” the possibility that some industries could be “completely obliterated,” and systems that might “keep the lights on while fixing [themselves] in the background.”

These comments do not establish what will happen. They identify a problem that practitioners are already trying to understand: how organizations retain control, judgment, and the capacity to learn when systems can adapt without continuous human intervention. The idea of self-healing systems raises a further question. The issue is not simply whether a system can diagnose and repair itself but how an organization knows what the system changed, when to intervene, and whether a correction at one level creates failure elsewhere.

Research should examine how organizations detect and recover from self-reinforcing system failures, how they preserve the capacity to intervene as systems adapt, how people remain meaningfully engaged with work increasingly coordinated by autonomous systems, and how professional judgment and accountability change as systems become more self-organizing. Systems thinking can help explain recurring patterns of coordination, breakdown, adaptation, and recovery (Checkland, 1981; Matook & Brown, 2017; Nan & Lu, 2014). The aim is not to predict a single future, but to help organizations recognize and shape plausible paths before those paths become difficult to change. These topics do not exhaust the IS research agenda; some consequential issues may not yet appear in practitioner accounts.

These topics bring us back to the motivation for this editorial. Practitioner concerns do more than identify problems organizations want solved. They can also signal phenomena that research has not yet fully explained. Across cybersecurity, value realization, governance, and resilience, executives are trying to understand where AI enters work, what authority it receives, who bears the costs of its use, and how organizations remain secure and able to adapt as their dependence on it grows. Some of these problems are already visible; others are only beginning to take shape. Staying close to practice gives IS researchers a better chance to see when familiar assumptions no longer fit and to study new arrangements as they develop, rather than reconstruct them after they have become established. Reconnecting with practice therefore means more than strengthening the implications section after a study is complete. It means allowing practitioner problems to help shape the questions we ask in the first place.

Conclusion

Practice is facing a more rapidly changing technology landscape. Past IS research helps us understand some of today's organizational reactions, but the problems continue to evolve. As a discipline, we are very good at conducting post-mortems of past events. If we take the opportunity to engage with organizations as they experience this rapid change, we can contribute a forward-looking perspective to both science and practice.

As we celebrate the 50th volume of *MIS Quarterly*, it is also worth remembering that engagement with practice has been part of the journal from its earliest years. The IT issues studies we revisit in this editorial are just a few examples of how the journal has used practitioner concerns to identify important problems for IS research. Fifty years later, the technologies and many of the problems have changed, but the importance of that connection has not.

Without the organizational engagement that is so much a part of the history of *MIS Quarterly*, we risk missing the real problems organizations face. That engagement can also help demonstrate the broader value of our research at a time when public support for higher education can no longer be taken for granted (Unruh, 2025). We hope this editorial inspires researchers to reengage with organizations and advance our understanding of the issues they face now and will face in the coming years.

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